

Supporting Information

Forecasting Tail-Event Probabilities Across Equity-Market Sectors: Do Hawkes-Informed Graph Models Add Predictive Value?

This Supporting Information reports the data-construction audits, GJR-GARCH/EVT diagnostics, Hawkes lagged-excitation analysis, common-shock robustness, nested model-tuning and calibration design, five-seed neural diagnostics, graph placebos, component ablations, tail-event-definition robustness, forecast-comparison inference, calibration and decision-value checks, formal sector-by-horizon heterogeneity tests, and reproducibility audits supporting the main manuscript.

The primary outcome is a one-day conditional extreme downside event: the sector return falls below a rolling 2.5% downside threshold derived from a GJR-GARCH-filtered peaks-over-threshold EVT model. Legacy computational fields retain names such as `Crash_Main` and `Crash_0025` for backward compatibility, but the supporting text refers to these observations as tail events. The study evaluates the six primary sectors at 1-, 5-, 10-, and 22-trading-day horizons and, unless otherwise stated, compares forecasts over the locked Test period from 1 August 2024 to 31 July 2026.

The revised forecast pipeline separates TrainFit, TrainTune, Calibration, and Test. The study uses only TrainFit and TrainTune for model-selection decisions, refits the models on the complete Train period, calibrates probabilities on the separate 1 August 2022–31 July 2024 Calibration period, and accesses the Test period only after freezing all pre-Test choices. The final post-processing stage uses saved predictions and does not retrain the model.

SI A. Data construction, sample flow, and sector eligibility

The firm-level panel contains 107,930 stock-day observations for 52 stocks across 11 NSE sectors and 2,488 market trading dates from 1 August 2016 to 31 July 2026. The study calculates sector returns using previous-trading-day market-capitalisation weights only when the constituent-coverage rules are satisfied. The primary analysis retains six sectors with at least three observed constituents and at least 90% eligible-day coverage. The study does not reconstruct firms absent from the supplied historical data; it excludes firms with insufficient price, shares, volume, or sector-classification information. This filtering is necessary for consistent estimation but may introduce sample-selection or survivorship bias if excluded firms differ systematically from retained firms. The study sources the firm-level NSE data from African Markets. The study does not redistribute raw proprietary observations in the reproducibility package.

Table S1. Data audit, sector eligibility, and primary-sample composition

Panel A. Firm-level data audit

Audit item	Value
Raw rows	107930
Unique stocks	52
Unique sectors	11
Unique parsed dates	2488
Minimum date	2016-08-01
Maximum date	2026-07-31
Duplicate stock-date rows	0

Panel B. Sector eligibility and primary-sample composition

Sector	Stocks	Eligible coverage (%)	Primary	Eligibility decision
Agriculture	6	67.04	No	Eligible sector-day coverage below 90%
Automobiles and Accessories	1	0.00	No	Fewer than 3 constituent stocks
Banking	10	99.88	Yes	Primary
Commercial and services	11	98.87	Yes	Primary
Construction and Allied	2	27.37	No	Fewer than 3 constituent stocks
Energy and Petroleum	4	99.00	Yes	Primary
Insurance	6	99.92	Yes	Primary
Investment	3	99.60	Yes	Primary
Investment Services	1	0.00	No	Fewer than 3 constituent stocks
Manufacturing and Allied	7	92.81	Yes	Primary
Telecommunication and Technology	1	0.00	No	Fewer than 3 constituent stocks

Note. A primary sector requires at least three observed constituent stocks and at least 90% eligible-day coverage. A sector-day is eligible only when at least two valid firm returns are available, and constituent-return coverage is at least two-thirds.

The data audit confirms a clean firm-level panel with no duplicate stock-date observations over the 2016–2026 study period. The primary analysis retains six of the 11 NSE sectors because they satisfy the pre-specified constituent-depth and coverage requirements. The excluded sectors fail mainly because of limited constituent representation or insufficient eligible-day coverage. The resulting six-sector sample therefore prioritises consistent longitudinal coverage, although excluding incompletely observed firms and sectors may introduce sample-selection or survivorship bias.

Table S2. Revised temporal sample flow and return-construction robustness

Panel A. Temporal sample flow

Phase	Calendar rows	Labelled sector-days	Current-day tail events	Event rate (%)
TrainFit	6,738	3,567	104	2.92
TrainTune	2,244	2,213	37	1.67
Calibration	2,976	2,933	73	2.49
Test	2,970	2,968	66	2.22

Note. The phase counts above describe current-day 2.5% tail-event labels in the R handoff. Forecast-target counts differ slightly because future-horizon targets apply split-aware censoring. In the final locked Test evaluation, the 1-, 5-, 10-, and 22-day targets contain 66, 304, 581, and 1,139 events, respectively.

Panel B. Return-construction robustness

Sector	Robustness	Common labels	Label agreement (%)	Event Jaccard	Return corr.	Threshold corr.
Commercial and services	Equal weight	1,960	96.07	0.105	0.598	-0.061
Insurance	Equal weight	1,986	96.98	0.294	0.783	0.586
Investment	Equal weight	1,978	95.75	0.125	0.419	0.149
Banking	Equal weight	1,985	98.19	0.446	0.872	0.871
Energy and Petroleum	Equal weight	1,963	97.55	0.273	0.819	0.734
Manufacturing and Allied	Equal weight	1,809	96.41	0.145	0.453	0.049
Energy and Petroleum	Strict liquidity	1,487	98.86	0.595	0.970	0.900

Sector	Robustness	Common labels	Label agreement (%)	Event Jaccard	Return corr.	Threshold corr.
Banking	Strict liquidity	1,636	99.39	0.767	0.993	0.973
Investment	Strict liquidity	1,391	97.48	0.364	0.949	0.661
Insurance	Strict liquidity	1,364	98.39	0.522	0.980	0.792
Commercial and services	Strict liquidity	1,217	98.19	0.450	0.987	0.834
Manufacturing and Allied	Strict liquidity	1,060	98.68	0.576	0.984	0.488

Note. Equal-weight robustness uses the same matched firm-day support as the primary lagged-market-capitalisation portfolio. Strict-liquidity robustness requires positive volume and excludes zero-return observations. Overall label agreement is high, but event-level Jaccard overlap is lower for equal weighting, indicating that portfolio weighting can change which specific days qualify as tail events.

Tail-event incidence remains low across the chronological phases, with 66 current-day events in the locked Test period. Return-construction robustness is stronger under the stricter-liquidity specification, which produces very high return correlations and generally greater event overlap with the primary market-capitalisation-weighted construction. Equal weighting maintains high overall label agreement but yields lower event-level Jaccard overlap, showing that portfolio weighting can change which specific sector-days the model classifies as tail events.

SI B. Dynamic volatility and EVT diagnostics

The primary tail-event rule combines one-step GJR-GARCH conditional moments with a peaks-over-threshold Generalised Pareto Distribution fitted to adverse standardised innovations. GJR-GARCH uses Student-t innovations, a 500-observation initial history, and 22-trading-day recursive refits. EVT uses an expanding history strictly before the labelled date, a 90th-percentile adverse-shock threshold, a minimum of 40 exceedances, and 22-day refits. The primary tail probability is 2.5%, with 1% and 5% definitions retained for robustness.

Table S3. EVT parameter summaries, tail-event incidence, and realised event magnitudes

Panel A. Sector-level EVT parameter summaries

Sector	Refits	Mean u	Mean exceed. (%)	Mean scale	Mean shape	Shape SD	Boundary	Non-conv.
Banking	91	1.108	9.98	0.575	0.232	0.043	0	0
Commercial and services	90	1.163	9.98	0.592	0.034	0.164	0	0
Energy and Petroleum	90	1.116	9.98	0.562	0.156	0.039	0	0
Insurance	91	1.047	9.98	0.627	0.125	0.055	0	0
Investment	90	0.966	9.98	0.825	0.019	0.065	0	0
Manufacturing and Allied	83	0.991	9.98	0.672	0.021	0.075	0	0

Panel B. Tail-event incidence by probability level

Tail probability	Observed labels	Tail events	Event rate (%)
1%	11,681	116	0.993
2.5%	11,681	280	2.397
5%	11,681	582	4.982

Panel C. Realised tail-event magnitudes by sample phase

Phase	Events	Mean return (%)	Median return (%)	5th pct return (%)	Worst return (%)	Mean threshold shortfall (pp)
Calibration	73	-4.35	-4.09	-7.69	-9.56	1.14
Test	66	-4.17	-3.81	-7.78	-14.43	1.07
TrainFit	104	-4.44	-3.84	-7.98	-28.59	1.33
TrainTune	37	-4.49	-4.23	-7.90	-9.25	1.53

The sector-level EVT estimates are numerically stable, with no boundary or non-convergence cases across the primary refits. Tail-event incidence increases as expected from the 1% to 5% probability definitions, while the primary 2.5% rule identifies 280 events. These events are economically material across all sample phases: in the locked Test period, the mean realised return is -4.17% and the median is -3.81% . The results therefore support using the EVT rule to identify sparse but substantial downside events without interpreting every observation as a conventional market crash.

Table S4. EVT threshold stability, tail-fit diagnostics, parameter uncertainty, and refit reliability

Panel A. POT-threshold stability and mean-residual-life diagnostics

POT threshold percentile	Diagnostic fits	Valid fits	Valid (%)	Mean exceedances	Mean residual life	Mean shape
85.0	142	142	100.0	224.2	0.676	0.121
87.5	142	142	100.0	186.5	0.688	0.120
90.0	142	142	100.0	149.4	0.716	0.099
92.5	142	136	95.8	112.1	0.729	0.109
95.0	142	118	83.1	74.6	0.745	0.157

Panel B. PIT and tail-QQ diagnostics

Sector	Diagnostic refits	PIT KS $p < 0.05$	Mean PIT KS p	Mean tail-QQ corr.
Banking	24	0	0.515	0.969
Commercial and services	24	5	0.233	0.824
Energy and Petroleum	24	0	0.625	0.993
Insurance	24	0	0.837	0.990
Investment	24	0	0.905	0.993
Manufacturing and Allied	22	0	0.800	0.995

Panel C. GPD parameter uncertainty

Sector	Diagnostic refits	Hessian OK	Mean shape SE	Mean 95% shape-CI width
Banking	24	24	0.095	0.373
Commercial and services	24	24	0.064	0.251
Energy and Petroleum	24	24	0.104	0.408
Insurance	24	24	0.111	0.434
Investment	24	24	0.093	0.364
Manufacturing and Allied	22	22	0.104	0.409

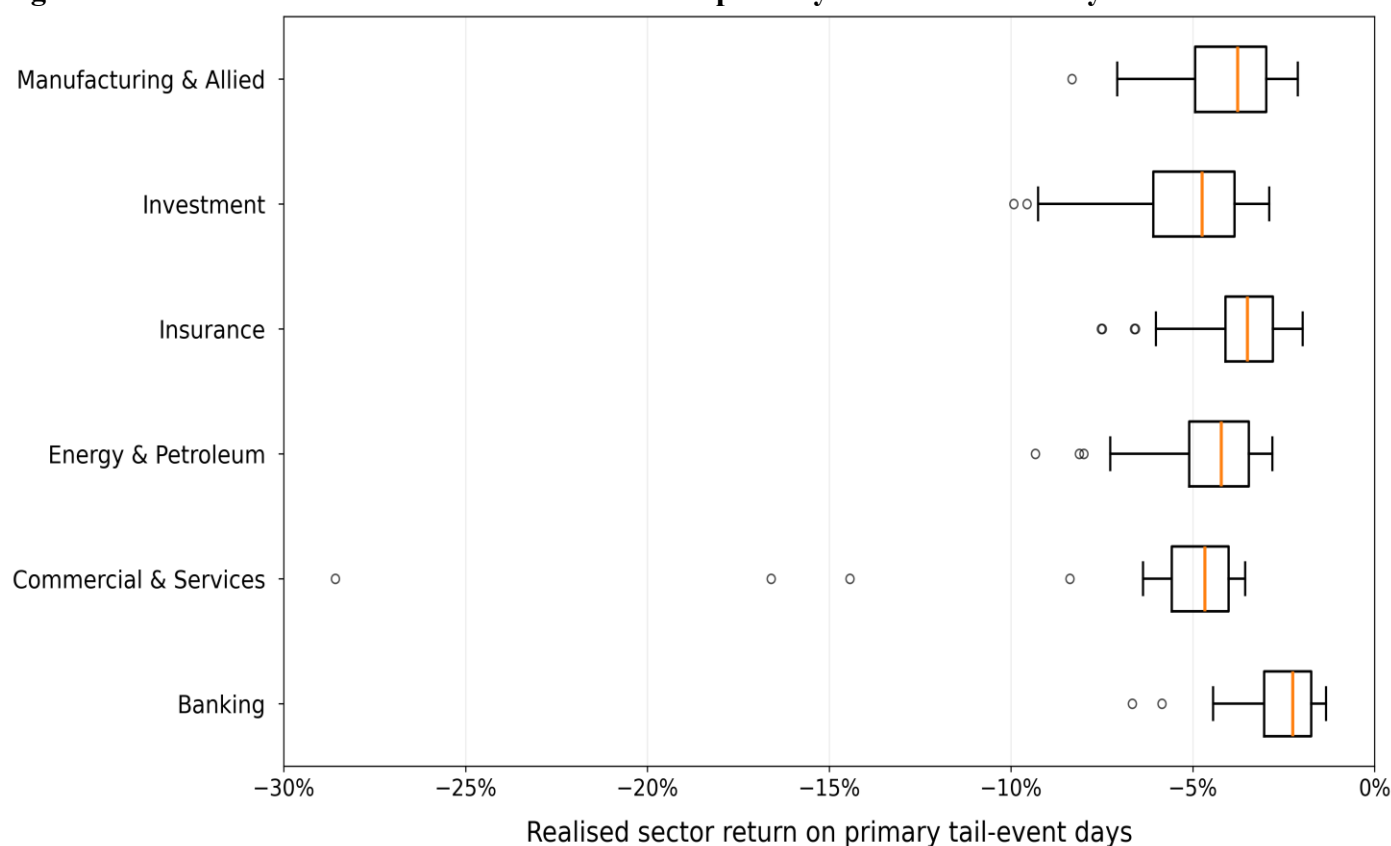
Panel D. EVT refit reliability

Sector	Scheduled	Successful	Failed	Fallback carry-forwards
Banking	91	91	0	0
Commercial and services	90	90	0	0
Energy and Petroleum	90	90	0	0
Insurance	91	91	0	0
Investment	90	90	0	0
Manufacturing and Allied	83	83	0	0

Note. The 85%, 87.5%, and 90% threshold diagnostics converge in all 142 sampled refits; valid-fit rates decline to 95.8% at 92.5% and 83.1% at 95% as exceedance counts fall. The 90% choice therefore balances tail focus with estimation adequacy. PIT and tail-QQ diagnostics are generally strong; Commercial and Services is the least uniform, with five of 24 sampled PIT tests rejecting at 5% and a mean tail-QQ correlation of 0.824. All 535 scheduled primary EVT refits succeed, with no fallback carry-forwards or boundary fits.

Taken together, the diagnostics support the 90% POT threshold as a defensible primary specification. Estimation remains fully valid through the 90% threshold, while reliability declines at more extreme thresholds as exceedance counts fall. Tail fit is generally satisfactory across sectors, although Commercial & Services shows comparatively weaker diagnostics. Parameter uncertainty is measurable but estimable in all diagnostic refits, and the absence of failed or fallback primary EVT refits indicates strong computational reliability.

Figure S1. Distribution of realised sector returns on primary 2.5% tail-event days

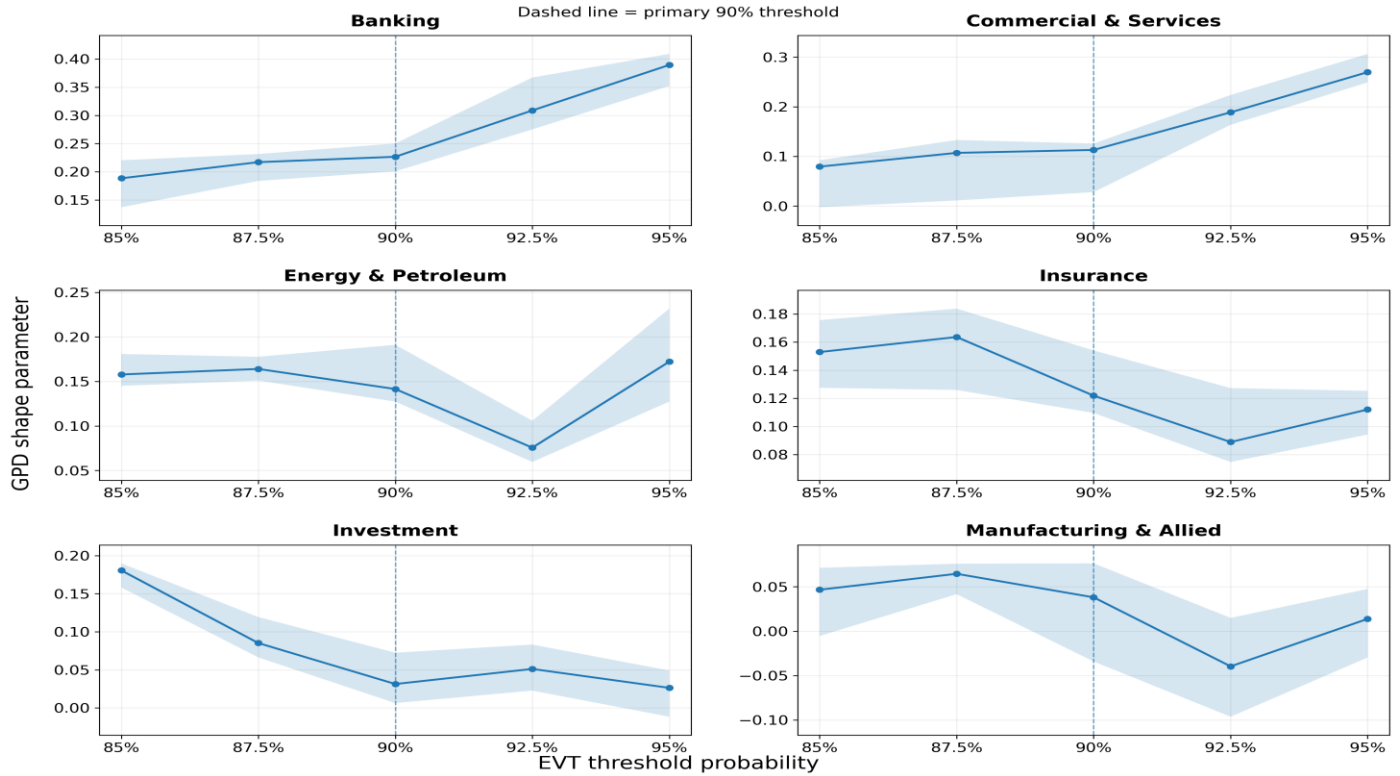


Note. More negative values indicate larger adverse returns. The box represents the interquartile range, the line inside the box is the median, whiskers extend to $1.5 \times IQR$, and points beyond the whiskers are outliers.

Tail-event returns are concentrated in economically meaningful negative ranges across all six sectors, although their severity varies by sector. Banking shows comparatively smaller losses, whereas Investment, Commercial & Services, and Manufacturing & Allied exhibit more negative central

outcomes. Commercial & Services also contains the most extreme downside observations, including losses approaching 30%. Overall, the distribution confirms that the identified tail events represent material downside episodes rather than minor daily fluctuations.

Figure S2. GPD shape-parameter stability across candidate POT thresholds



Note. Points show the median GPD shape estimate; the shaded band is the interquartile range (25th–75th percentiles). The dashed line marks the primary 90% POT threshold.

The GPD shape estimates are generally stable around the primary 90% POT threshold, while greater variation emerges at the more extreme 92.5% and 95% thresholds. Banking and Commercial & Services show increasing tail-shape estimates at higher thresholds, whereas Investment declines and Manufacturing & Allied becomes more variable. Overall, the figure supports the 90% threshold as a reasonable balance between tail sensitivity and parameter stability across sectors.

SI C. Hawkes lagged-excitation estimation, stability, and common-shock robustness

The Hawkes analysis estimates lagged predictive dependence among sector tail events, not contemporaneous causality. For sector i on day t , the Bernoulli event probability is linked to a non-negative intensity that combines a baseline term, self-excitation, and lagged excitation states from the six sectors. The event state uses observations through $t-1$ only, so same-day sector events cannot excite one another. The cross-sector L1 penalty is applied only to off-diagonal excitation coefficients; self-excitation is unpenalised. The study estimates parameters by bounded L-BFGS-B optimisation with non-negative coefficients and five multistarts for final fits. Candidate half-lives are 1, 2, 5, 10, and 22 trading days; the full six-sector model has 42 parameters. Stability selection uses 100 block subsamples for the primary lambda-min specification, 70% subsample fraction, 60-day blocks, and a 60% retention rule. The discrete branching matrix $A/(1-\delta)$ must have a spectral radius below 0.98.

Table S5. Hawkes persistence, penalty selection, and stability sensitivity

Panel A. Hawkes half-life selection

Half-life	Decay	Train log-lik.	Parameters	AIC	BIC	Delta BIC
5	0.8706	-637.704	42	1359.408	1639.219	0.000

Half-life	Decay	Train log-lik.	Parameters	AIC	BIC	Delta BIC
10	0.9330	-637.758	42	1359.516	1639.326	0.108
2	0.7071	-638.552	42	1361.104	1640.915	1.696
22	0.9690	-640.383	42	1364.765	1644.576	5.357
1	0.5000	-641.122	42	1366.244	1646.055	6.836

Panel B. L1-penalty selection

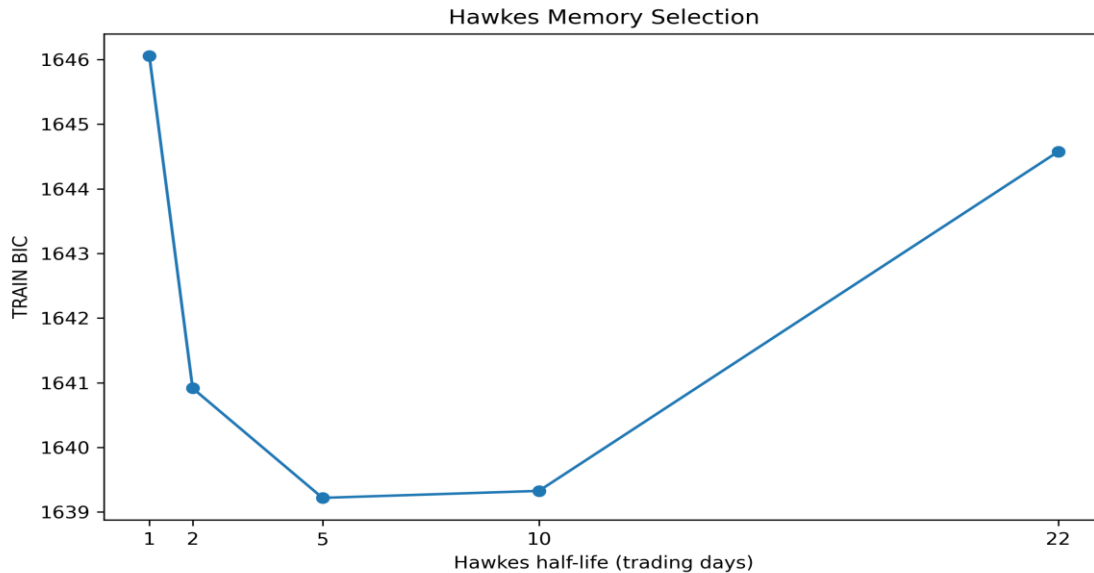
Half-life	Lambda-min	Min CV Log Score	SE at min	Lambda-1SE	Primary penalty	Robustness penalty
2.0	0.0	0.3744	0.0478	0.5	0.0	0.5
5.0	0.1	0.3797	0.0408	0.5	0.1	0.5
10.0	0.2	0.3787	0.0399	0.5	0.2	0.5

Panel C. Primary lagged-excitation link stability and coefficient uncertainty

Primary lagged-excitation link	Selection freq.	Mean coeff.	Empirical 95% interval	>=70%	>=80%	>=90%
Banking -> Insurance	0.94	0.0222	[0.0000, 0.0367]	Yes	Yes	Yes
Insurance -> Manufacturing and Allied	0.99	0.0231	[0.0058, 0.0407]	Yes	Yes	Yes
Investment -> Banking	0.62	0.0078	[0.0000, 0.0283]	No	No	No

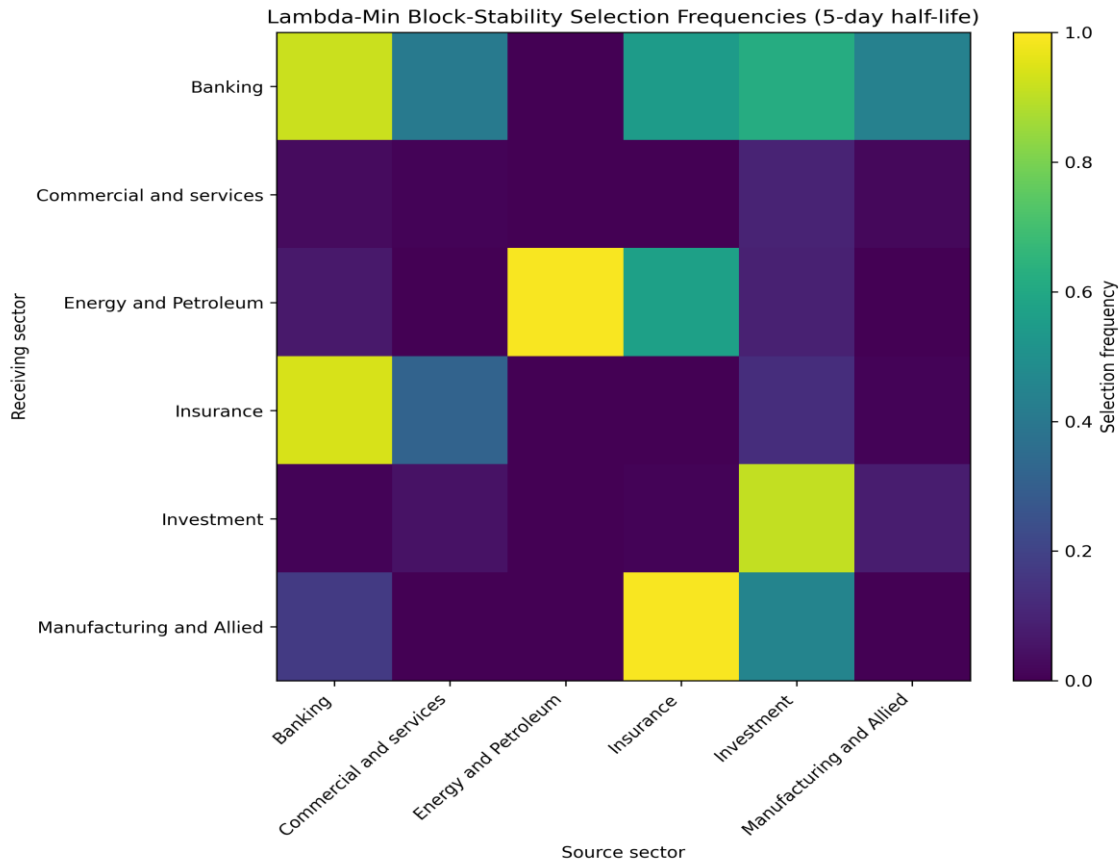
The five-day Hawkes specification provides the best overall persistence fit, although the 10-day alternative is nearly equivalent. Stability selection identifies three primary lagged-excitation links, but their robustness differs materially. Banking → Insurance and Insurance → Manufacturing & Allied remain stable under stricter retention thresholds, whereas Investment → Banking is retained only at the primary 60% cutoff. The results therefore indicate a sparse and uneven pattern of cross-sector predictive dependence rather than a broadly connected excitation network.

Figure S3. Hawkes half-life selection by training-sample BIC.



The training-sample BIC reaches its minimum at the five-day Hawkes half-life, while the 10-day specification produces only a marginally higher value. This pattern indicates that the most plausible excitation persistence lies within approximately five to ten trading days, while much shorter or longer half-lives fit the data less well.

Figure S4. Primary five-day Hawkes lagged-excitation stability heatmap



The stability heatmap identifies strong cross-sector lagged excitation in only a few source–receiver pairs. The highest off-diagonal selection frequencies correspond to Investment → Banking, Banking → Insurance, and Insurance → Manufacturing & Allied, while most other cross-sector relationships remain weak. The resulting network is therefore sparse rather than broadly interconnected.

Table S6. Common-shock-controlled Hawkes robustness, same-day treatment, and standalone forecast skill

Panel A. Common-shock robustness of primary lagged-excitation links

Lagged-excitation link	Primary freq.	Common-shock freq.	Survives 60%	70%	80%	90%
Investment -> Banking	0.62	0.32	No	No	No	No
Banking -> Insurance	0.94	0.60	Yes	No	No	No
Insurance -> Manufacturing and Allied	0.99	0.93	Yes	Yes	Yes	Yes

Panel B. Market-state coefficient estimates

Recipient sector	Mean market-state coefficient	Empirical 95% interval
Banking	0.0214	[0.0000, 0.0579]
Commercial and services	0.0039	[0.0000, 0.0248]
Energy and Petroleum	0.0188	[0.0000, 0.0403]
Insurance	0.0480	[0.0080, 0.0789]
Investment	0.0091	[0.0000, 0.0284]
Manufacturing and Allied	0.0224	[0.0000, 0.0436]

Panel C. Same-day excitation and forecast-timing audit

Audit item	Treatment
Sector excitation state entering $\lambda_{i,t}$	Uses sector events through $t-1$ only
Market common-shock state entering $\lambda_{i,t}$	Uses market events through $t-1$ only
Same-day sector-to-sector excitation	Not permitted
Same-day market-to-sector excitation	Not permitted
State used for forecast $t+1$ from origin t	Post-origin state includes events observed at t , then forecasts $t+1$ onward

Panel D. Standalone Hawkes forecast performance

Horizon	N	Events	Brier	Log Score	Brier skill vs expanding history
1	2960	66	0.022041	0.110594	-0.0086
5	2928	304	0.096426	0.349639	-0.0249
10	2890	581	0.168185	0.528822	-0.0272
22	2832	1139	0.259077	0.722658	-0.0524

Controlling for the NSE-wide market tail-event state materially weakens two of the three primary lagged-excitation links. Investment $\rightarrow$ Banking no longer meets the 60% retention threshold, Banking $\rightarrow$ Insurance survives only at 60%, whereas Insurance $\rightarrow$ Manufacturing & Allied remains highly stable through the 90% cutoff. The market-state effect is clearest for Insurance, whose empirical 95% interval excludes zero. The timing audit confirms that same-day excitation is not permitted, and the negative Brier skill of standalone Hawkes forecasts at all four horizons indicates that the network is more informative as a structural conditioning mechanism than as a standalone probability forecaster.

SI D. Nested forecast design, model capacity, calibration, and five-seed neural diagnostics

The revised forecasting design separates model selection from probability calibration. TrainFit spans 1 August 2016-29 January 2021 and estimates candidate models. TrainTune spans 1 February 2021-29 July 2022 and is used only to select neural epochs, XGBoost boosting rounds, and the ridge-logit penalty. After freezing these choices, the study refits each model from scratch on the full Train period through 29 July 2022. The distinct Calibration period from 1 August 2022 to 31 July 2024 is then used only for Platt or temperature/bias calibration. The locked Test period begins 1 August 2024. Split-aware future-event targets stop at each phase boundary, so no outcome from a later phase enters an earlier-phase target.

Table S7. Nested model-selection design, model capacity, and calibration audit

Panel A. Nested sample sizes and XGBoost boosting rounds

Horizon	XGB rounds	TrainFit N	TrainTune N	Full Train N	Calibration N	Test N
1	555	3468	2178	5652	2889	2960
5	359	3196	2089	5315	2738	2928
10	1	3017	2020	5094	2605	2890
22	1	2792	1919	4816	2402	2832

Panel B. Neural-model parameter counts.

Neural architecture	Trainable parameters	Invariant across seeds
DynamicHawkesGraphTransformer	75,960	Yes
LSTM_Survival	45,622	Yes

Neural architecture	Trainable parameters	Invariant across seeds
NoGraphTransformer	75,960	Yes
RandomGraphTransformer	75,960	Yes
StaticHawkesGraphTransformer	75,960	Yes
TemporalTransformer_Survival	66,310	Yes

Panel C. Neural epoch selection across five seeds

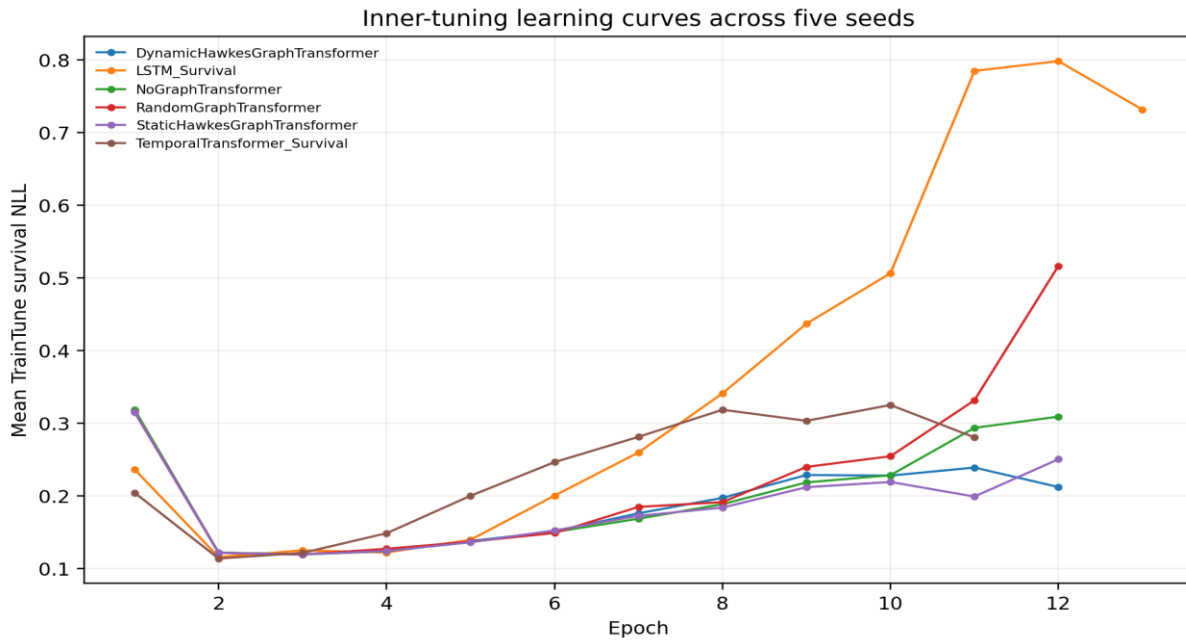
Neural model	Min selected epoch	Max selected epoch	Mean selected epoch
DynamicHawkesGraphTransformer	2	4	2.60
LSTM_Survival	2	5	3.00
NoGraphTransformer	2	4	2.60
RandomGraphTransformer	2	4	2.60
StaticHawkesGraphTransformer	2	4	2.60
TemporalTransformer_Survival	2	3	2.20

Panel D. Dynamic ridge-logit penalty selection

Horizon	Dynamic ridge-logit lambda	Best TrainTune Log Score	Upper grid boundary
1	10.0	0.089195	Yes
5	10.0	0.306100	Yes
10	10.0	0.474336	Yes
22	10.0	0.680243	Yes

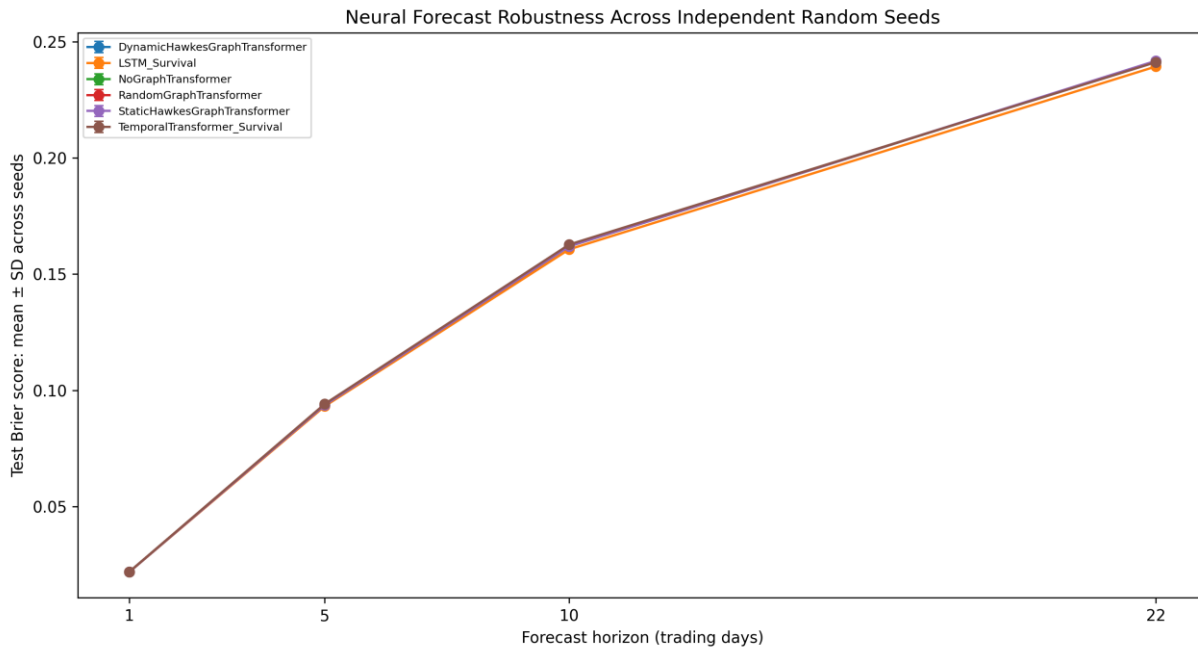
Note. The final comparison estimates 70 parameters for the dynamic ridge-logit. The tuning procedure selects $\lambda = 10$, the upper boundary of the pre-specified grid, at every horizon, and the study reports this choice transparently without expanding the grid after Test evaluation. The 2022–2024 Calibration period is used only for post-estimation probability calibration and does not enter epoch, boosting-round, or penalty selection.

The nested design provides substantial pre-Test samples at every horizon while keeping tuning, refitting, calibration, and final evaluation chronologically separated. Neural architectures contain between 45,622 and 75,960 trainable parameters, yet TrainTune selects only a small number of training epochs across all five seeds, indicating limited training duration. XGBoost also becomes highly parsimonious at the 10- and 22-day horizons, selecting one boosting round. The dynamic ridge-logit consistently selects the strongest pre-specified penalty, reinforcing its role as a comparatively parsimonious benchmark while preserving a fully leakage-controlled evaluation design.

Figure S5. Inner-tuning learning curves across five neural seeds

Note. Curves show mean TrainTune survival negative log-likelihood by epoch for each architecture. Selected epochs are generally early, indicating that the TrainTune criterion favoured limited training duration across architectures.

The TrainTune loss curves generally reach their minima within the first few epochs, after which validation loss either flattens or deteriorates for most architectures. Prolonged training increases losses sharply for the LSTM and random-graph Transformer, making this pattern especially pronounced for these models. The results therefore support the early stopping choices and indicate that additional epochs would have increased overfitting rather than improved out-of-sample tuning performance.

Figure S6. Five-seed neural Test Brier Score mean and standard deviation.

Test Brier Scores are very similar across the five independent neural seeds at each forecast horizon, indicating low sensitivity to random initialisation. Performance differences across architectures are also small relative to the increase in forecast difficulty as the horizon lengthens. Overall, the results suggest that unstable seed-specific outcomes do not drive the neural-model comparisons.

SI E. Locked Test forecast performance, graph placebos, and forecast-comparison inference

The final forecast evaluation uses proper probability losses as primary criteria. Table S8 combines the principal machine-learning, neural, econometric, Hawkes, and historical benchmarks using the same locked Test observations. The final inference layer then applies paired moving-block bootstrap inference at block lengths of 10, 22, and 44 trading days, a stationary-bootstrap sensitivity with mean block length 22, and Holm adjustment across the pre-specified focal comparisons.

Table S8. Primary locked Test model comparison and Brier skill

Panel A. Primary locked Test forecast performance

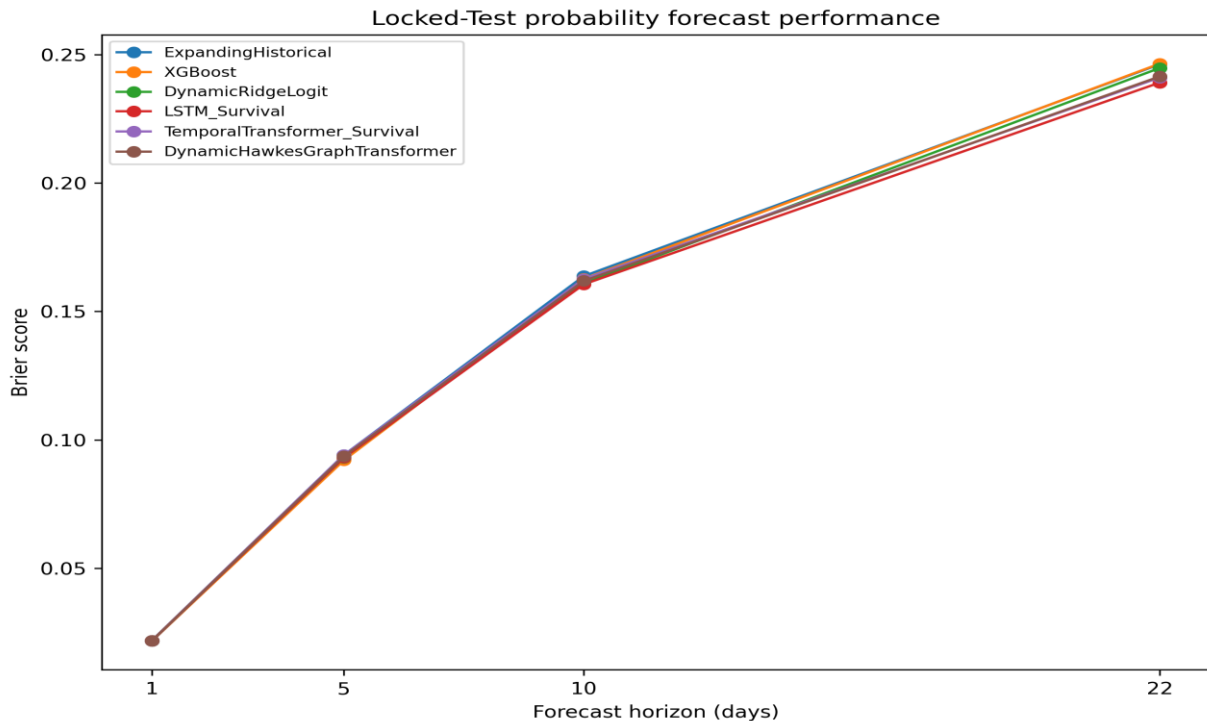
H	Model	N	Events	Brier	Log Score	PR-AUC	ROC-AUC	Brier skill (%)
1	XGBoost	2960	66	0.021755	0.105586	0.0369	0.6374	0.45
1	LSTM Survival	2960	66	0.021800	0.106849	0.0320	0.5247	0.25
1	Dynamic ridge-logit	2960	66	0.021807	0.106998	0.0248	0.5289	0.21
1	Dynamic Hawkes Graph Transformer	2960	66	0.021822	0.107280	0.0267	0.5195	0.15
1	No-Graph Transformer	2960	66	0.021822	0.107289	0.0267	0.5180	0.14
1	Expanding Historical	2960	66	0.021853	0.107971	0.0200	0.4244	0.00
1	Stable Hawkes	2960	66	0.022041	0.110594	0.0220	0.4779	-0.86
5	XGBoost	2928	304	0.092088	0.327891	0.1553	0.6147	2.12
5	Dynamic ridge-logit	2928	304	0.093036	0.333318	0.1108	0.5305	1.12
5	LSTM Survival	2928	304	0.093107	0.333747	0.1285	0.5379	1.04
5	Dynamic Hawkes Graph Transformer	2928	304	0.093635	0.336195	0.1136	0.4994	0.48
5	No-Graph Transformer	2928	304	0.093648	0.336253	0.1128	0.4981	0.47
5	Expanding Historical	2928	304	0.094088	0.338503	0.0919	0.4288	0.00
5	Stable Hawkes	2928	304	0.096426	0.349639	0.0969	0.4676	-2.49
10	LSTM Survival	2890	581	0.160552	0.501662	0.2382	0.5339	1.94
10	Dynamic ridge-logit	2890	581	0.160875	0.502800	0.2161	0.5301	1.74
10	Dynamic Hawkes Graph Transformer	2890	581	0.161879	0.505445	0.2170	0.4956	1.13
10	No-Graph Transformer	2890	581	0.161910	0.505529	0.2169	0.4938	1.11
10	XGBoost	2890	581	0.162717	0.507740	0.2134	0.5056	0.62
10	Expanding Historical	2890	581	0.163728	0.511004	0.1797	0.4268	0.00
10	Stable Hawkes	2890	581	0.168185	0.528822	0.1923	0.4724	-2.72
22	LSTM Survival	2832	1139	0.239059	0.671076	0.4582	0.5452	2.89
22	Dynamic Hawkes Graph Transformer	2832	1139	0.241443	0.675960	0.4575	0.5103	1.92
22	No-Graph Transformer	2832	1139	0.241498	0.676072	0.4569	0.5070	1.90
22	Dynamic ridge-logit	2832	1139	0.244738	0.685571	0.4609	0.5454	0.58
22	Expanding Historical	2832	1139	0.246176	0.685901	0.3859	0.4431	0.00
22	XGBoost	2832	1139	0.246426	0.685995	0.4022	0.5000	-0.10
22	Stable Hawkes	2832	1139	0.259077	0.722658	0.4046	0.4718	-5.24

Note. XGBoost has the lowest Brier Score at 1 and 5 days; LSTM Survival has the lowest Brier Score at 10 and 22 days. Stable Hawkes underperforms the expanding historical benchmark at every horizon. Positive Brier skill indicates lower Brier loss than the expanding historical reference.

Panel B. Brier skill and moving-block-bootstrap confidence intervals

Model	Horizon	Brier skill (%)	95% MBB22 CI (%)
XGBoost	1	0.45	[-0.14, 0.98]
XGBoost	5	2.12	[0.09, 5.32]
XGBoost	10	0.62	[-2.02, 2.75]
XGBoost	22	-0.10	[-5.97, 4.08]
LSTM Survival	1	0.25	[0.13, 0.38]
LSTM Survival	5	1.04	[0.48, 1.65]
LSTM Survival	10	1.94	[0.84, 3.04]
LSTM Survival	22	2.89	[0.75, 4.74]
Dynamic ridge-logit	1	0.21	[0.07, 0.40]
Dynamic ridge-logit	5	1.12	[0.08, 2.77]
Dynamic ridge-logit	10	1.74	[-0.60, 5.27]
Dynamic ridge-logit	22	0.58	[-7.06, 8.51]
Dynamic Hawkes Graph Transformer	1	0.15	[-0.13, 0.36]
Dynamic Hawkes Graph Transformer	5	0.48	[-0.75, 1.31]
Dynamic Hawkes Graph Transformer	10	1.13	[-0.49, 2.84]
Dynamic Hawkes Graph Transformer	22	1.92	[-0.83, 4.40]

Forecast performance varies materially by horizon. XGBoost records the lowest Brier Score at 1 and 5 days, whereas the LSTM performs best at 10 and 22 days. Relative to the expanding historical benchmark, the largest Brier-skill gains are generally modest. However, LSTM improvements at 1 and 10 days remain statistically significant after Holm adjustment for both Brier and Log Score. The dynamic ridge-logit remains competitive despite its much lower model complexity, while the standalone Hawkes model underperforms the historical benchmark at every horizon. Overall, the results favour horizon-specific model performance rather than a single uniformly dominant forecasting architecture.

Figure S7. Main locked Test Brier Score by forecast horizon.

Brier loss increases with forecast horizon for all models, reflecting the greater difficulty of predicting whether a tail event will occur over longer windows. Model differences are small at 1 and 5 days but become more visible at 10 and 22 days. XGBoost performs best at the shorter horizons, whereas the LSTM achieves the lowest Brier Score at 10 and 22 days. The figure therefore reinforces the absence of a single model that dominates across all forecast horizons.

Table S9. Graph placebo design and dynamic-versus-no-graph inference

Panel A. Graph-placebo specifications

Graph variant	Description	Cross edges	Row normalised	Graph-state features
No Graph	Zero econometric prior; graph attention remains data-learned; graph-state summaries are zero.	0	NO	ZERO
Random Static Graph	Random directed prior with Hawkes cross-edge density; same Train-derived magnitude scale.	3	NO	YES
Static Hawkes Graph	Time-invariant stability-selected Hawkes coefficient prior; same Train-derived magnitude scale.	3	NO	YES
Dynamic Hawkes Graph	Time-varying stability-selected Hawkes excitation prior; absolute excitation magnitude preserved.	3	NO	YES

Panel B. Dynamic-versus-no-graph Brier-loss comparisons

Horizon	Dynamic - no-graph Brier	95% MBB22 CI	Raw p	Holm p	Holm sig.
1	-0.00000049	[-0.00000092, -0.00000002]	0.0319	0.3513	No
5	-0.00001321	[-0.00002238, -0.00000167]	0.0160	0.1916	No
10	-0.00003080	[-0.00005979, 0.00000377]	0.0699	0.6287	No
22	-0.00005391	[-0.00011600, 0.00000579]	0.0818	0.6547	No

Note. Negative differences favour the Dynamic Hawkes Graph Transformer. The point estimates are extremely small. Although the unadjusted 95% intervals at 1 and 5 days exclude zero, none of the dynamic-versus-no-graph Brier comparisons remains significant after Holm adjustment. Block-length and stationary-bootstrap sensitivity led to the same substantive conclusion: Hawkes-informed graph conditioning provides numerically small incremental forecast gains that do not remain statistically significant after multiplicity adjustment.

The placebo comparison shows that adding the Hawkes-informed dynamic graph changes Brier loss only marginally relative to the no-graph Transformer. Although the unadjusted 1- and 5-day intervals favour the dynamic graph, these gains do not remain significant after Holm correction, and the 10- and 22-day intervals include zero. The evidence therefore indicates that Hawkes-informed graph conditioning adds little incremental predictive value beyond the underlying Transformer architecture.

SI F. Component ablations and tail-event-definition robustness

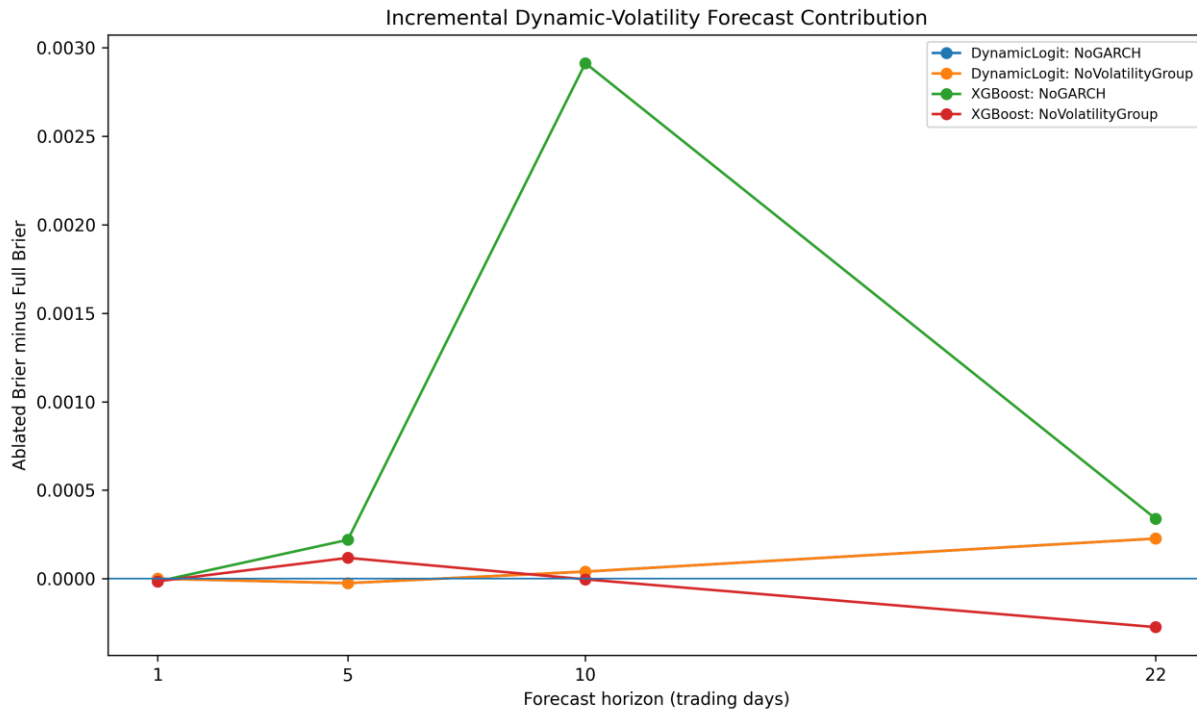
Component ablations assess whether GJR-GARCH volatility and the broader volatility block add forecast information after the model retains the remaining predictors. Tail-event-definition robustness compares the primary volatility-adjusted 2.5% EVT rule with a sector-specific fixed 2.5% return threshold. Because event prevalence and difficulty differ across definitions, comparisons emphasise Brier skill relative to each definition's own strictly past-based expanding historical benchmark rather than raw Brier scores across different outcomes.

Table S10. Incremental forecast contribution of volatility information

Model	Ablation	H	Full - ablation Brier	95% MBB22 CI	Bootstrap P(Full better)
XGBoost	NoGARCH	1	0.000015	[-0.000010, 0.000044]	0.110
XGBoost	NoGARCH	5	-0.000222	[-0.000943, 0.000507]	0.726
XGBoost	NoGARCH	10	-0.002871	[-0.006862, 0.000557]	0.954
XGBoost	NoGARCH	22	0.000029	[-0.004270, 0.003805]	0.592
XGBoost	NoVolatilityGroup	1	0.000014	[-0.000024, 0.000059]	0.240
XGBoost	NoVolatilityGroup	5	-0.000109	[-0.000678, 0.000383]	0.642
XGBoost	NoVolatilityGroup	10	0.000010	[-0.002107, 0.001795]	0.592
XGBoost	NoVolatilityGroup	22	0.000510	[-0.003020, 0.003949]	0.396
Dynamic ridge-logit	NoGARCH	1	-0.000000	[-0.000000, 0.000000]	0.636
Dynamic ridge-logit	NoGARCH	5	0.000025	[-0.000109, 0.000155]	0.380
Dynamic ridge-logit	NoGARCH	10	-0.000034	[-0.000354, 0.000299]	0.552
Dynamic ridge-logit	NoGARCH	22	0.000081	[-0.004332, 0.002584]	0.546
Dynamic ridge-logit	NoVolatilityGroup	1	-0.000000	[-0.000000, 0.000000]	0.640
Dynamic ridge-logit	NoVolatilityGroup	5	0.000025	[-0.000090, 0.000161]	0.348
Dynamic ridge-logit	NoVolatilityGroup	10	-0.000034	[-0.000431, 0.000291]	0.590
Dynamic ridge-logit	NoVolatilityGroup	22	0.000081	[-0.003626, 0.002849]	0.528

Note. NoGARCH removes the GJR-GARCH conditional-volatility feature, whereas NoVolatilityGroup removes the broader volatility-feature block. Negative Full-minus-ablation Brier differences favour the Full specification. Boot. P(Full better) is the proportion of bootstrap replications in which the Full specification produces lower Brier loss than the corresponding ablation.

The contribution of volatility information is small and varies across models and forecast horizons. The largest numerical improvement occurs for XGBoost at 10 days when the GJR-GARCH feature is retained (difference = -0.002871), but its 95% confidence interval includes zero. The remaining contrasts are smaller and similarly uncertain. Overall, the ablation results do not establish a consistent reduction in Brier loss from retaining either the GJR-GARCH feature or the broader volatility block.

Figure S8. Incremental contribution of volatility information

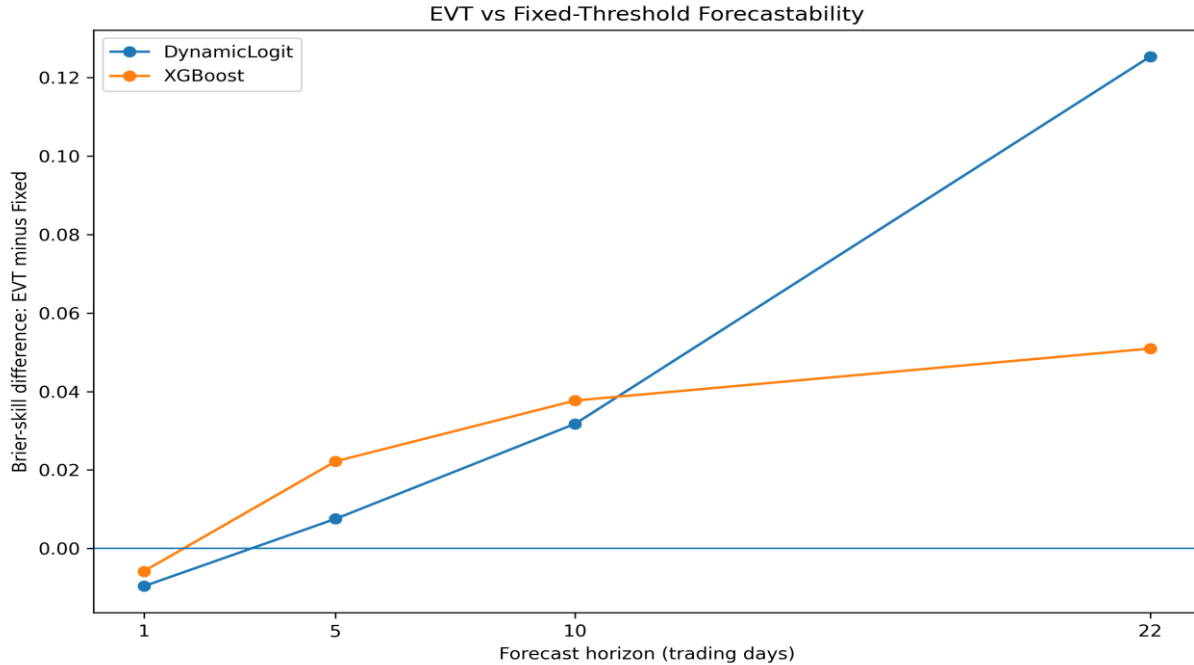
The incremental contribution of volatility information is modest at most horizons and differs by model specification. XGBoost records the clearest numerical gain at the 10-day horizon when the model retains the GJR-GARCH feature, whereas the broader volatility block adds little further improvement. For the dynamic ridge-logit, effects remain close to zero across horizons. The figure therefore reinforces that volatility information adds selective rather than uniform predictive value.

Table S11. Volatility-adjusted EVT versus fixed-threshold Brier-skill comparison

Model	H	EVT skill (%)	Fixed skill (%)	EVT - fixed (pp)	95% MBB22 CI (pp)	Bootstrap P(EVT higher)
XGBoost	1	0.33	0.91	-0.58	[-1.95, 0.31]	0.086
XGBoost	5	2.58	0.37	2.22	[-2.28, 5.94]	0.834
XGBoost	10	0.59	-3.18	3.77	[-4.11, 10.21]	0.836
XGBoost	22	0.59	-4.50	5.09	[-3.86, 13.47]	0.816
Dynamic ridge-logit	1	0.21	1.18	-0.96	[-2.60, 0.33]	0.078
Dynamic ridge-logit	5	1.12	0.37	0.75	[-2.34, 3.17]	0.690
Dynamic ridge-logit	10	1.73	-1.45	3.17	[-2.56, 7.81]	0.860
Dynamic ridge-logit	22	-1.11	-13.65	12.54	[0.42, 22.41]	0.976

Note. The study calculates each Brier-skill measure relative to the strictly past-based expanding historical benchmark for the corresponding tail-event definition. EVT – fixed reports the difference in Brier skill between the volatility-adjusted EVT and fixed-threshold definitions; positive values favour EVT. Boot. P(EVT higher) is the proportion of bootstrap replications in which EVT produces higher definition-specific Brier skill than the fixed-threshold definition.

The relative advantage of the volatility-adjusted EVT definition varies by model and horizon. XGBoost favours EVT numerically at 5, 10, and 22 days, but all corresponding 95% confidence intervals include zero. For the dynamic ridge-logit, the clearest evidence occurs at 22 days, where EVT improves Brier skill by 12.54 percentage points, and the 95% interval excludes zero. Overall, the results indicate that volatility-adjusted EVT identification can improve forecast skill in selected settings, but the benefit is not uniform across models or horizons.

Figure S9. Brier-skill comparison: 2.5% EVT versus fixed 2.5% return threshold

The relative forecast advantage of the volatility-adjusted EVT definition increases with horizon for both models. XGBoost shows positive EVT-minus-fixed Brier-skill differences from 5 days onward, while the dynamic ridge-logit records a much larger gain at 22 days. The figure therefore suggests that volatility-adjusted EVT identification becomes more useful at longer horizons, although the statistical evidence is strongest only for the 22-day dynamic ridge-logit comparison.

SI G. Calibration, practical decision value, and formal sector-horizon heterogeneity

The study evaluates calibration on the locked Test period after estimating all calibration parameters on the separate 2022-2024 Calibration sample. Reliability diagrams use ten probability bins with moving-block-bootstrap uncertainty bands. Brier decomposition separates reliability, resolution, and outcome uncertainty. Decision-curve analysis compares each model's net benefit with treat-all and treat-none strategies over thresholds from 0.01 to 0.50. These analyses assess calibration and potential decision usefulness after model selection and do not affect model choice or tuning.

Table S12. Calibration, Brier decomposition, and decision-value summary

Panel A. Calibration-in-the-large and calibration slope

Model	H	Mean predicted - observed	Calibration intercept	Calibration slope
XGBoost	1	-0.0058	0.013	0.925
XGBoost	5	-0.0190	0.334	1.046
XGBoost	10	0.0468	5.598	6.287
XGBoost	22	0.0774	-0.293	1.262
LSTM Survival	1	0.0026	3.000	1.850
LSTM Survival	5	0.0152	1.674	1.916
LSTM Survival	10	0.0186	0.553	1.526
LSTM Survival	22	0.0128	0.261	1.917
Dynamic ridge-logit	1	0.0027	20.000	6.493
Dynamic ridge-logit	5	-0.0058	-0.705	0.653

Model	H	Mean predicted - observed	Calibration intercept	Calibration slope
Dynamic ridge-logit	10	-0.0193	-0.559	0.543
Dynamic ridge-logit	22	-0.0730	-0.026	0.509
Dynamic Hawkes Graph Transformer	1	0.0047	-0.033	1.045
Dynamic Hawkes Graph Transformer	5	0.0242	-1.325	0.433
Dynamic Hawkes Graph Transformer	10	0.0356	-0.640	0.632
Dynamic Hawkes Graph Transformer	22	0.0374	0.242	2.632

Panel B. Brier-score decomposition

Model	H	Reliability	Resolution	Uncertainty	Actual Brier
XGBoost	1	0.000118	0.000151	0.021800	0.021755
LSTM Survival	1	0.000117	0.000117	0.021800	0.021800
Dynamic ridge-logit	1	0.000037	0.000030	0.021800	0.021807
Dynamic Hawkes Graph Transformer	1	0.000083	0.000062	0.021800	0.021822
Expanding Historical	1	0.000140	0.000085	0.021800	0.021853
XGBoost	5	0.001235	0.001869	0.093045	0.092088
LSTM Survival	5	0.001128	0.001073	0.093045	0.093107
Dynamic ridge-logit	5	0.000438	0.000438	0.093045	0.093036
Dynamic Hawkes Graph Transformer	5	0.001658	0.001088	0.093045	0.093635
Expanding Historical	5	0.003183	0.002113	0.093045	0.094088
XGBoost	10	0.008522	0.006385	0.160622	0.162717
LSTM Survival	10	0.002188	0.002305	0.160622	0.160552
Dynamic ridge-logit	10	0.001839	0.001572	0.160622	0.160875
Dynamic Hawkes Graph Transformer	10	0.004253	0.003025	0.160622	0.161879
Expanding Historical	10	0.010420	0.007180	0.160622	0.163728
XGBoost	22	0.025395	0.019403	0.240433	0.246426
LSTM Survival	22	0.006819	0.008284	0.240433	0.239059
Dynamic ridge-logit	22	0.011786	0.006912	0.240433	0.244738
Dynamic Hawkes Graph Transformer	22	0.010616	0.009634	0.240433	0.241443
Expanding Historical	22	0.020514	0.014074	0.240433	0.246176

Panel C. Decision-curve net-benefit summary

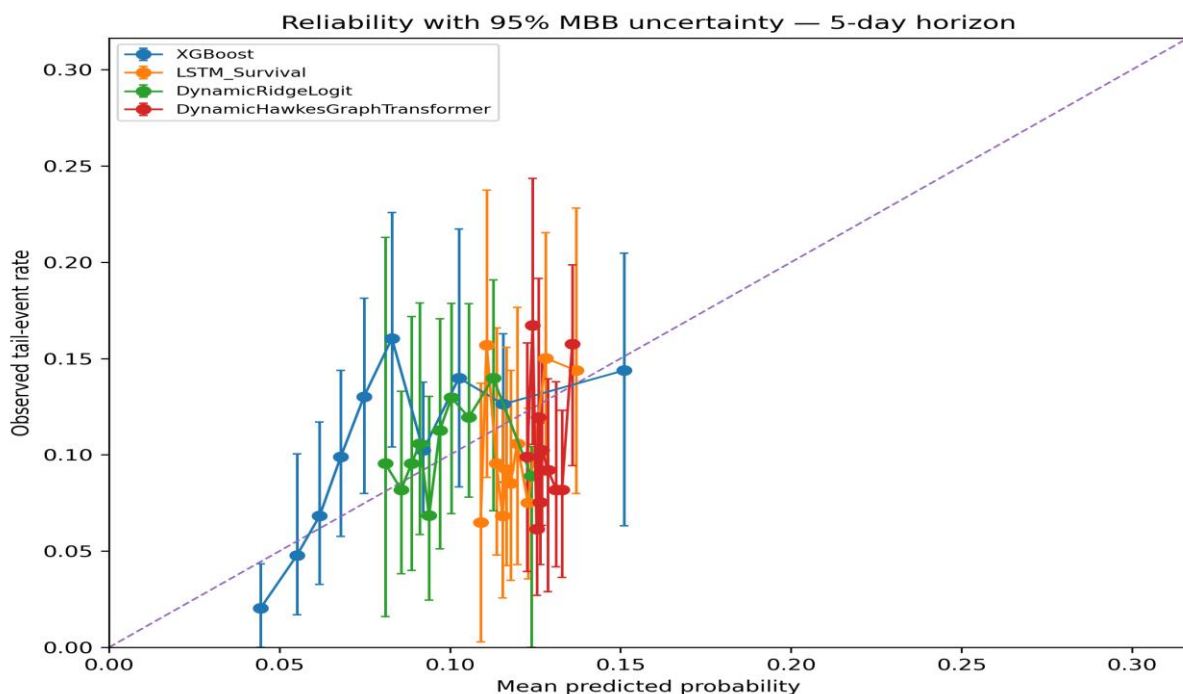
Model	H	Thresholds better than both	Min threshold	Max threshold	Max NB advantage
XGBoost	1	4	0.01	0.05	0.0020
XGBoost	5	27	0.03	0.29	0.0085
XGBoost	10	4	0.26	0.29	0.0014
XGBoost	22	0	—	—	0.0000
LSTM Survival	1	1	0.03	0.03	0.0003
LSTM Survival	5	2	0.12	0.13	0.0028

Model	H	Thresholds better than both	Min threshold	Max threshold	Max NB advantage
LSTM Survival	10	4	0.22	0.27	0.0075
LSTM Survival	22	7	0.39	0.49	0.0406
Dynamic ridge-logit	1	0	—	—	0.0000
Dynamic ridge-logit	5	3	0.08	0.11	0.0031
Dynamic ridge-logit	10	4	0.13	0.21	0.0157
Dynamic ridge-logit	22	15	0.19	0.50	0.0306
Dynamic Hawkes Graph Transformer	1	0	—	—	0.0000
Dynamic Hawkes Graph Transformer	5	0	—	—	0.0000
Dynamic Hawkes Graph Transformer	10	1	0.25	0.25	0.0013
Dynamic Hawkes Graph Transformer	22	2	0.45	0.46	0.0239

Note. The study evaluates calibration statistics on the locked Test sample after estimating calibration parameters exclusively on the separate 2022–2024 Calibration period. In Panel C, "Thresholds better than both" counts probability thresholds at which the model has greater net benefit than both treat-all and treat-none strategies; Max NB advantage is the largest net-benefit improvement over those reference strategies.

Calibration quality varies across models and forecast horizons rather than remaining uniformly close to ideal values. The Brier decomposition nevertheless shows that the fitted models generally reduce the reliability component relative to the expanding historical benchmark, particularly at the longer horizons. Decision value is also strongly horizon- and threshold-dependent: XGBoost has its broadest useful range at five days, whereas the LSTM and dynamic ridge-logit provide wider positive net-benefit regions at 22 days. The Dynamic Hawkes Graph Transformer offers only narrow decision-value regions at the longer horizons and none at 1 or 5 days. Overall, the results show that practical forecast usefulness depends on both the model, the forecast horizon, and the decision threshold rather than on probability accuracy alone.

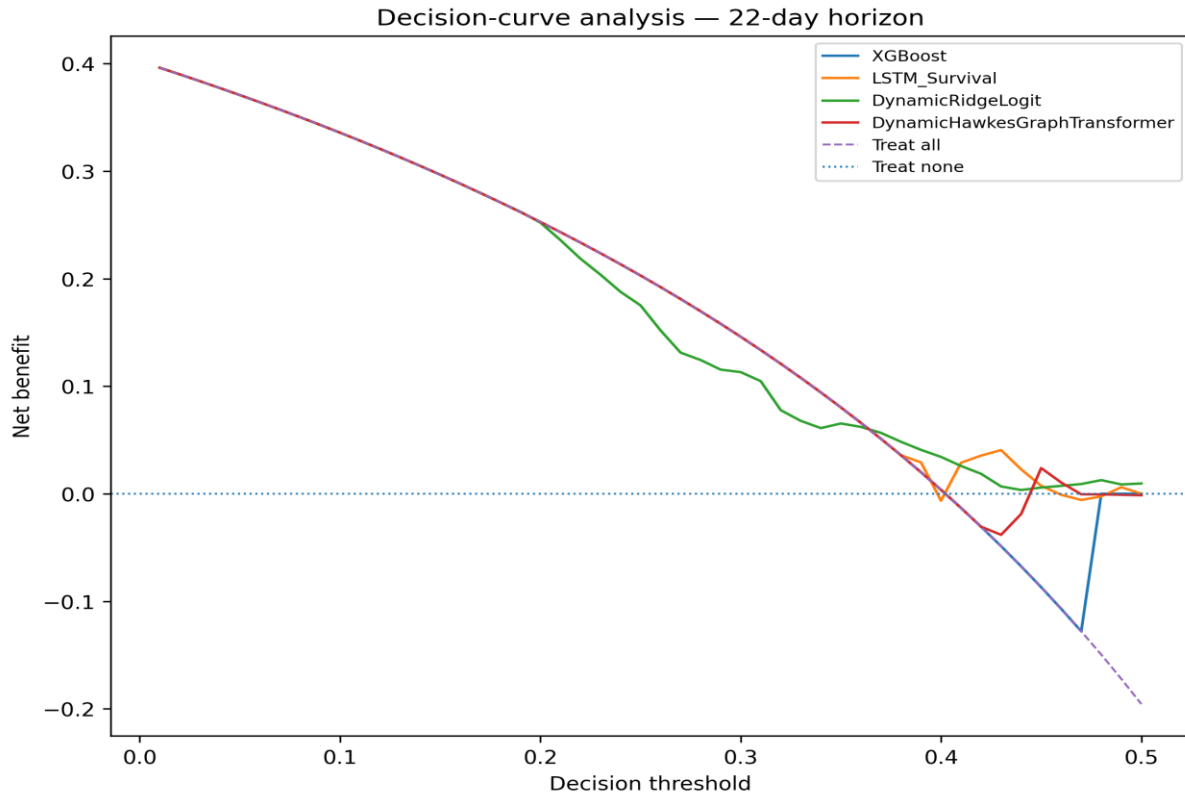
Figure S10. Reliability diagram at the 5-day horizon



At the five-day horizon, calibration varies across models and probability ranges. XGBoost generally tracks the 45-degree calibration line more closely across a wider range of predicted probabilities, while

the LSTM, dynamic ridge-logit, and Dynamic Hawkes Graph Transformer show larger local departures. The bootstrap uncertainty intervals are nevertheless wide and frequently overlap the ideal-calibration line, indicating considerable sampling uncertainty. Overall, the figure suggests reasonable but imperfect five-day calibration rather than uniformly well-calibrated probabilities across models.

Figure S11. Decision curves at the 22-day horizon

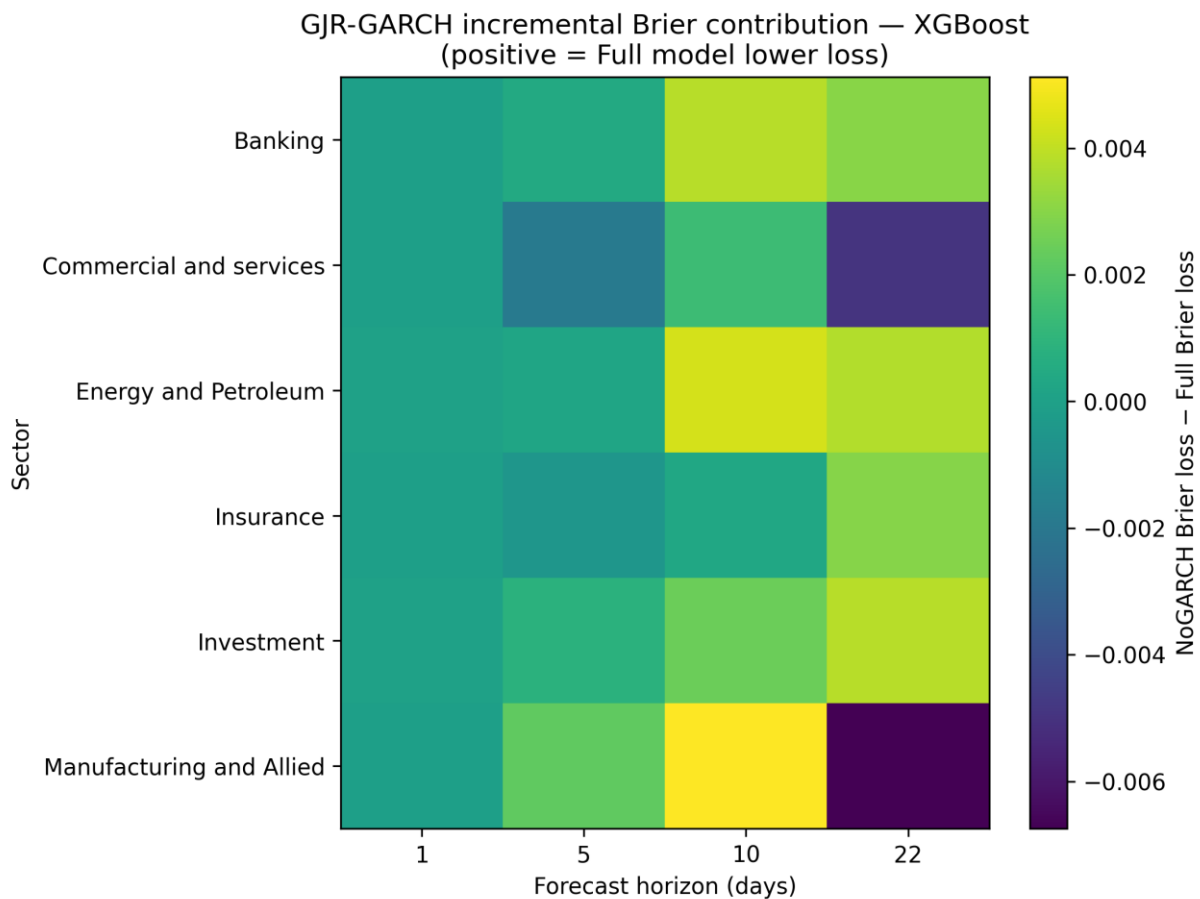


At the 22-day horizon, decision value varies substantially across models and probability thresholds. The dynamic ridge-logit provides the broadest range of positive net benefit relative to both treat-all and treat-none strategies, while the LSTM becomes useful mainly at higher thresholds. The Dynamic Hawkes Graph Transformer offers positive net benefit only over a narrow threshold range, and XGBoost does not outperform both reference strategies. The figure therefore shows that practical usefulness at longer horizons depends strongly on the decision threshold and model choice.

Table S13. Formal sector-by-horizon heterogeneity tests

Model	Loss	Q statistic	Raw p	Holm p	Holm significant
XGBoost	Brier	0.084071	0.242	0.483	No
XGBoost	LogScore	0.505334	0.333	0.667	No
Dynamic ridge-logit	Brier	0.044495	0.449	1.000	No
Dynamic ridge-logit	LogScore	0.234099	0.453	1.000	No

Note. The omnibus tests assess whether the incremental contribution of GJR-GARCH information differs across the 24 sector-by-horizon cells. Holm-adjusted *p*-values control for multiplicity across the reported model-loss comparisons.

Figure S12. Sector-by-horizon GJR-GARCH contribution for XGBoost

The incremental contribution of the GJR-GARCH feature varies noticeably across sector-horizon cells. Positive contributions are most visible at the 10- and 22-day horizons for sectors such as Banking, Energy & Petroleum, Insurance, and Investment. In contrast, Commercial & Services and Manufacturing & Allied show negative contributions at some longer horizons. These contrasts show descriptive variation across sectors and horizons, but the non-significant omnibus tests do not establish statistically significant heterogeneity.

SI H. Feature-importance audit and reproducibility checks

XGBoost gain importance is reported only as a descriptive audit of which predictors drive split improvements. It does not establish causality or indispensability. The final code also records all split definitions, selected boosting rounds and neural epochs, calibration parameters, model parameter counts, random seeds, software versions, saved predictions, and integrity checks. The final inference script reads frozen saved predictions and performs no model retraining.

Table S14. Feature-importance and reproducibility audit

Panel A. Top ten aggregated XGBoost gain-importance features by forecast horizon

Horizon	Rank	Original feature (code field)	Gain share (%)
1	1	SectorReturn_Model	4.56
1	2	StdInnovation	4.37
1	3	EVT_nhist	3.94
1	4	EVT_scale	3.86
1	5	MarketZeroVolumeShare	3.75
1	6	EVT_u	3.59

Horizon	Rank	Original feature (code field)	Gain share (%)
1	7	EVT_pu	3.55
1	8	GARCH_Sigma	2.81
1	9	EVT_nexc	2.80
1	10	GARCH_Mu	2.71
5	1	SectorReturn_Model	5.22
5	2	GARCH_Sigma	5.15
5	3	StdInnovation	4.72
5	4	EVT_u	4.57
5	5	EVT_pu	4.38
5	6	EVT_scale	4.37
5	7	GARCH_Mu	4.14
5	8	EVT_nhist	3.76
5	9	GARCH_Realized	3.05
5	10	MarketZeroVolumeShare	2.96
10	1	EVT_u	20.67
10	2	GARCH_Mu	17.21
10	3	MarketAmihudILLIQ_Median	16.65
10	4	MarketCapKES_Total	9.26
10	5	MarketReturn_MCW	8.21
10	6	CrossSectorReturnDispersion	6.98
10	7	MarketZeroVolumeShare	6.79
10	8	ReturnCoverage	5.38
10	9	NSectorsWithReturn	3.75
10	10	GARCH_Realized	3.13
22	1	EVT_u	21.57
22	2	EVT_scale	16.30
22	3	GARCH_Mu	13.34
22	4	EVT_shape	11.90
22	5	EVT_pu	9.20
22	6	GARCH_Realized	7.46
22	7	GARCH_Sigma	5.61
22	8	StdInnovation	5.46
22	9	EVT_nexc	5.05
22	10	__EVENT_HISTORY__	3.90

Note. Gain share measures each code field's relative contribution to XGBoost split improvements. It is descriptive and does not establish causality or predictor indispensability. The study retains the full feature-importance outputs in the computational output bundle.

The predictors receiving the greatest XGBoost gain importance change substantially with forecast horizon. At 1 and 5 days, sector returns, standardised innovations, GJR-GARCH volatility, and EVT variables appear frequently among the highest-ranked features. At 10 days, market-liquidity, market-return, market-capitalisation, and cross-sector dispersion measures become more prominent, while the 22-day model places particularly high importance on EVT threshold and tail-parameter information. The

rankings therefore suggest that the information XGBoost uses shifts from shorter-run sector and volatility conditions toward broader market and tail-state information as the forecast horizon lengthens.

Panel B. Reproducibility and integrity audit

Reproducibility item	Recorded implementation
Raw sample	107,930 stock-day rows; 52 stocks; 11 sectors; 2,488 market dates
Primary sectors	6 sectors passing constituent-depth and $\geq 90\%$ eligible-day coverage
Primary EVT refits	535/535 successful; 0 failed; 0 fallback carry-forwards; 0 boundary fits
Hawkes primary stability	100 block subsamples; 60-day blocks; 60% primary cutoff; 70/80/90% sensitivity
Neural stochastic audit	6 neural architectures $\times$ 5 fixed seeds; parameter counts recorded
Forecast chronology	TrainFit $\rightarrow$ TrainTune $\rightarrow$ full-Train refit $\rightarrow$ Calibration $\rightarrow$ locked Test
Final inference	Moving-block lengths of 10, 22, and 44 trading days plus stationary-bootstrap sensitivity; 500 replications
Multiplicity	Holm-adjusted focal forecast comparisons and sector-by-horizon omnibus tests
Final post-processing	All 12 integrity checks passed; no model retraining.
Raw-data redistribution	Not included; raw firm-level data remain subject to source access terms

Note. Raw proprietary firm-level observations are not redistributed and remain subject to source-access terms. The primary-output integrity comparison confirms that the final R code reproduces the frozen market-capitalisation-weighted construction. All 27,368 reference rows match, with zero primary-event or threshold mismatches and a maximum absolute sector-return difference of 4.996×10^{-16} .

The reproducibility audit confirms that the analytical workflow is chronologically separated, computationally traceable, and internally consistent. The study records all primary EVT refits, Hawkes stability checks, and neural seed checks, performs forecast inference only after freezing the locked Test predictions, and completes all 12 final post-processing integrity checks without retraining any models. The zero mismatch in the primary R construction further confirms that the additional diagnostics and robustness analyses did not alter the previously frozen primary sector-return and tail-event series.

Table S15. Computational environment and software versions

Analysis stage	Software/package	Version recorded	Recorded environment	Primary analytical role
R analysis	R	4.5.1 (2025-06-13 ucrt)	Windows 11 x64 (build 26200); x86_64-w64-mingw32/x64	GJR-GARCH, EVT, data construction, diagnostics and robustness analysis
R analysis	R script	3.4-final-audited	Same R environment	Frozen final R analysis pipeline
R package	rugarch	1.5-4	R 4.5.1	GJR-GARCH estimation and rolling forecasts
R package	ismev	1.43	R 4.5.1	Generalised Pareto distribution estimation
R package	FinTS	0.4-9	R 4.5.1	ARCH-LM diagnostic
R package	moments	0.14.1	R 4.5.1	Distributional diagnostics
R package	data.table	1.18.0	R 4.5.1	Panel construction, joins and tabulation
R package	dplyr	1.1.4	R 4.5.1	Data manipulation
R package	tidyr	1.3.2	R 4.5.1	Data reshaping
R package	lubridate	1.9.4	R 4.5.1	Date parsing and calendar handling
R package	xts	0.14.1	R 4.5.1	Time-indexed data support
R package	zoo	1.8-15	R 4.5.1	Ordered/time-series data support
R package	ggplot2	4.0.1	R 4.5.1	Diagnostic figures
R package	openxlsx	4.2.8.1	R 4.5.1	Excel output

Analysis stage	Software/package	Version recorded	Recorded environment	Primary analytical role
R package	arrow	25.0.1	R 4.5.1; loaded via namespace	Parquet output/serialisation
Python Phase 1	Python	3.13.15	Linux 6.6.122+ x86_64; glibc 2.39	Sparse Hawkes estimation, stability selection and common-shock robustness
Python Phase 1	Phase-1 script	1.3	Seed 20260901	Frozen final Hawkes pipeline
Python Phase 2.2	Python	3.13.15	Linux 6.6.122+ x86_64; glibc 2.39	Nested tuning, full-Train refit, calibration and five-seed forecast models
Python Phase 2.2	Phase-2.2 script	2.3	Seeds 20260901-20260905	Frozen five-seed forecasting pipeline
Python Phase 2.2	PyTorch	2.11.0+cu128	CUDA device	LSTM, temporal Transformer and graph Transformer estimation
Python Phase 3	Python	3.13.15	Linux 6.6.122+ x86_64; glibc 2.39	Ablations, dynamic ridge-logit and tail-event-definition robustness
Python Phase 3	Phase-3 script	3.2-audited	Seed 20260901	Frozen audited robustness pipeline
Python Phase 3	XGBoost	3.4.1	Python 3.13.15 environment	Gradient-boosted tree benchmark and ablation models
Final inference	Python	3.13.5	Linux 6.18.35 x86_64; glibc 2.41	Post-processing only; no model retraining
Final inference	Final-inference script	1.0-final-inference	Seed 20260912	Paired bootstrap inference, calibration, decision analysis, multiplicity and heterogeneity tests
Python dependency	NumPy / pandas / SciPy / scikit-learn / Matplotlib	The frozen run logs do not record the exact versions	Python analysis environments	The frozen scripts import these packages; the study does not infer their versions retrospectively.

Note. Versions shown are those explicitly recorded in the frozen R session information and Python run logs/metadata. The study reports Python dependency versions not captured in the frozen logs as unrecorded rather than inferring them retrospectively. Phase 2.2 used CUDA-enabled PyTorch. The final-inference stage used saved predictions only and performed no model retraining.

The study fully documents the computational environment across the R, Hawkes, forecasting, robustness, and final-inference stages, recording main software versions, operating environments, script versions, and random seeds for reproducibility. The forecasting models were estimated in a CUDA-enabled PyTorch environment, while the final inference stage was conducted separately using frozen predictions and involved no model retraining. Where the original run logs did not capture exact Python dependency versions, the study reports them as unrecorded rather than reconstructing them retrospectively. Overall, the environment record provides a traceable account of the software and execution settings underlying the reported results.

References

- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56. [https://doi.org/10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6)
- Bollerslev, T. (1986). Generalised autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1)
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
- Davison, A. C., & Smith, R. L. (1990). Models for exceedances over high thresholds. *Journal of the Royal Statistical Society: Series B (Methodological)*, 52(3), 393–442. <https://doi.org/10.1111/j.2517-6161.1990.tb01796.x>
- Garman, M. B., & Klass, M. J. (1980). On the estimation of security price volatilities from historical data. *The Journal of Business*, 53(1), 67–78. <https://doi.org/10.1086/296072>

- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. <https://doi.org/10.1111/j.1540-6261.1993.tb05128.x>
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. <https://doi.org/10.1198/016214506000001437>
- Hawkes, A. G. (1971). Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1), 83–90. <https://doi.org/10.1093/biomet/58.1.83>
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241. <https://doi.org/10.1214/aos/1176347265>
- Lim, B., Arik, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/10.1016/j.ijforecast.2021.03.012>
- McNeil, A. J., & Frey, R. (2000). estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300. [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8)
- Meinshausen, N., & Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(4), 417–473. <https://doi.org/10.1111/j.1467-9868.2010.00740.x>
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *The Journal of Business*, 53(1), 61–65. <https://doi.org/10.1086/296071>
- Politis, D. N., & Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313. <https://doi.org/10.1080/01621459.1994.10476870>
- Rogers, L. C. G., & Satchell, S. E. (1991). Estimating variance from high, low and closing prices. *The Annals of Applied Probability*, 1(4), 504–512. <https://doi.org/10.1214/aoap/1177005835>
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. <https://doi.org/10.1371/journal.pone.0118432>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. <https://arxiv.org/abs/1706.03762>